

FOUR-OUTCOME UNIVARIATE SUMMARY (1/2)

Predictor	Beta effect/q	Scale effect/q	Duration effect/q	Frequency effect/q
High-productivity hydrogeology	+0.9% / 0.576	-4.7% / 0.8	-1.8% / 0.273	-3.6% / 0.0377
Moderate-productivity hydrogeology	+2.2% / 0.0207	-24.9% / 0.0117	+0.1% / 0.921	-0.3% / 0.867
Low-productivity hydrogeology	-4.5% / 5.12e-05	+81.3% / 0.000356	-1.3% / 0.433	+2.3% / 0.175
Essentially no groundwater	+2.5% / 0.0338	-29.5% / 0.00406	+3.4% / 0.0333	+1.7% / 0.436
Intergranular-flow hydrogeology	+4.1% / 0.0155	-38.1% / 0.0014	+1.4% / 0.433	-7.4% / 3.55e-06
Fracture-flow hydrogeology	-3.0% / 0.00468	+50.0% / 0.000356	-2.9% / 0.0242	+3.3% / 0.0377
Static annual rainfall (1991-2020)	-4.8% / 0.0107	+62.7% / 0.0996	+4.7% / 0.122	-1.5% / 0.662
Static winter rainfall (1991-2020)	-12.3% / 0.00888	+254.2% / 0.0996	+19.9% / 0.0181	-3.4% / 0.662
Annual PRCPTOT	-2.6% / 0.185	+31.0% / 0.392	+4.1% / 0.0531	-2.6% / 0.238
Annual Rx1day	-1.3% / 0.708	+5.6% / 0.881	+5.4% / 0.161	-5.0% / 0.238
Annual Rx5day	-6.1% / 0.4	+94.1% / 0.49	+19.5% / 0.0265	-5.7% / 0.42
Annual SDII	-6.0% / 0.185	+121.8% / 0.327	+10.7% / 0.052	-3.9% / 0.42

Effects are unadjusted percentage associations per frozen meaningful increment; q is within-family BH FDR. Not causal.

FOUR-OUTCOME UNIVARIATE SUMMARY (2/2)

Predictor	Beta effect/q	Scale effect/q	Duration effect/q	Frequency effect/q
Annual R10mm days	-3.5% / 0.185	+48.9% / 0.327	+6.1% / 0.0265	-2.8% / 0.238
Annual R20mm days	-7.2% / 0.219	+119.9% / 0.405	+7.3% / 0.342	-2.1% / 0.7
Annual consecutive wet days	-5.1% / 0.0164	+77.1% / 0.0895	-1.3% / 0.539	-3.4% / 0.238
Annual consecutive dry days	+35.1% / 0.00635	-96.5% / 0.0125	-20.3% / 0.0265	+18.2% / 0.238
Urban land cover	+23.1% / 1.15e-07	-83.8% / 8.07e-05	-3.9% / 0.433	-8.1% / 0.175
Suburban land cover	+1.7% / 0.541	-18.5% / 0.554	+16.4% / 2.16e-08	+0.5% / 0.867
Continuous developed-land index	+15.4% / 1.8e-08	-72.3% / 7.83e-06	-8.6% / 0.0121	-6.9% / 0.058
Population density (2021)	+23.6% / 8.14e-08	-82.3% / 0.0014	+7.1% / 0.235	-9.8% / 0.103
Pre-1973 building share	+3.4% / 0.329	-29.4% / 0.393	-2.6% / 0.454	-3.1% / 0.662
Mean terrain slope	-4.5% / 0.0338	+36.1% / 0.324	-2.6% / 0.433	+1.8% / 0.662
Treatment capacity utilisation ratio	+3.4% / 0.336	-32.6% / 0.337	+14.8% / 6.2e-05	+7.2% / 0.0528
WWTW catchment area	+11.7% / 7.3e-06	-60.6% / 0.00407	-5.1% / 0.0325	-6.9% / 0.0241

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